

To what extent are EEG-based measures of awareness affected by neuromuscular blockade?

An experimental study

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The clinical problem is awareness without a behavioural signal

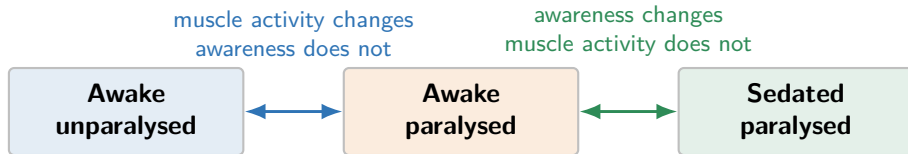
- ▶ Neuromuscular blocking agents improve intubation and surgical conditions.
- ▶ They also remove movement, the most obvious sign that a patient may still be aware.
- ▶ An EEG monitor can therefore be clinically useful only if it tracks awareness independently of muscle activity.

AAGA

Accidental awareness during general anaesthesia

Patient is aware
but classified as unaware

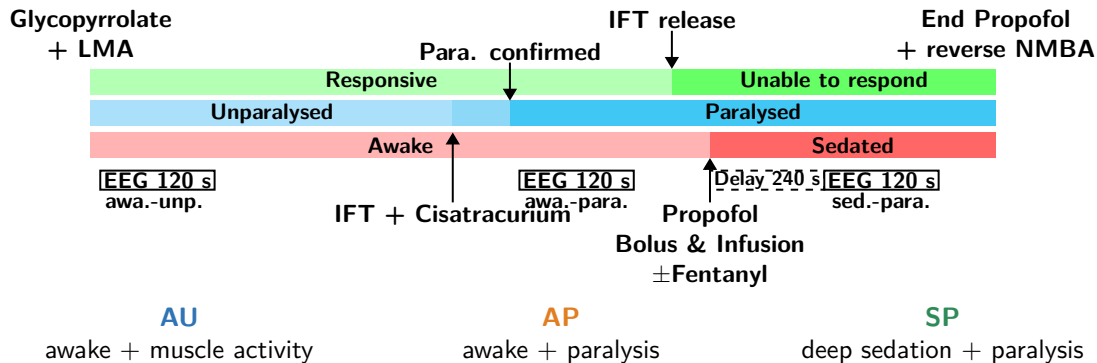
A valid marker must separate awareness from muscle activity



Minimal criterion

The measure should remain stable from **awake-unparalysed** to **awake-paralysed**, but change from **awake-paralysed** to **sedated-paralysed**.

The experiment creates exactly those two contrasts



Six volunteers and participant-held-out classification

- ▶ 6 healthy volunteers completed all three conditions
- ▶ 115 EEG channels, recorded at 5 kHz (downsampled to 250 Hz for analysis)
- ▶ 120 seconds selected per condition split into 5-second epochs with 50% overlap: 47 epochs per participant and condition

Leave-one-subject-out CV

Fit a linear SVC on five participants.

Predict the sixth participant and repeat until everyone has been held out.

AP is never used to fit the classifier.

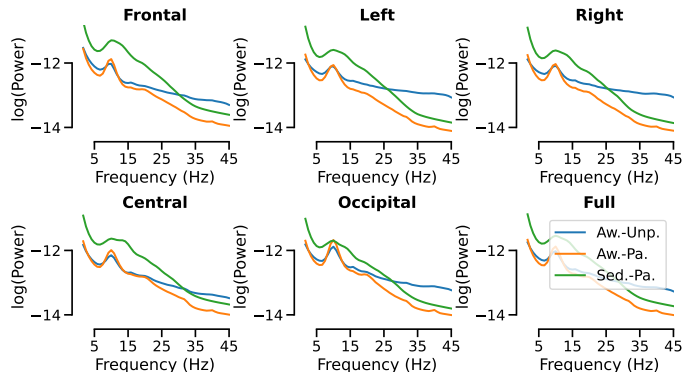
We tested measures commonly proposed as signatures of awareness

- ▶ Lempel–Ziv complexity
- ▶ Spectral exponent: low and high bands
- ▶ Directed transfer function outflow
- ▶ Alpha peak frequency
- ▶ Delta, theta, alpha, beta and gamma power
- ▶ Periodic and aperiodic spectral features
- ▶ Exploratory set of 55 measures

Question for every feature

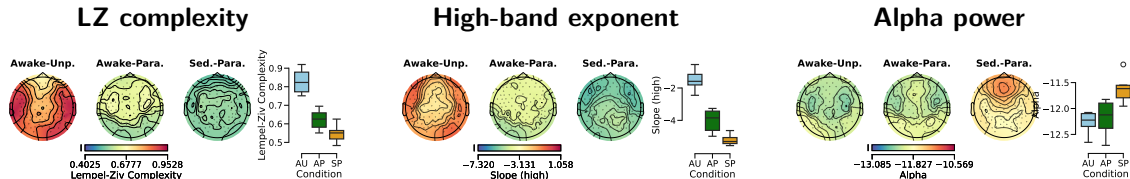
Does it recognise an aware person when paralysis removes the muscle signal?

Paralysis selectively removes high-frequency power



Below 15 Hz, the **awake-unparalysed** and **awake-paralysed** states are similar. Above 15 Hz, **awake-paralysed** shifts towards **sedated-paralysed**, especially at peripheral regions.

Several established measures move with paralysis, not awareness



LZ complexity and the high-band exponent show profound AU–AP changes; alpha is comparatively preserved across the two aware states.

Standard validation: awake versus sedated

Each fold trains on five participants and predicts AU and SP epochs from the held-out participant.

LZ complexity

True state	SP	1.00	0.00
	AU	0.02	0.98
		Una.	Awa.
		Predicted label	

99%

AU–SP epochs correct

Gamma power

True state	SP	0.96	0.04
	AU	0.00	1.00
		Una.	Awa.
		Predicted label	

98%

AU–SP epochs correct

Alpha power

True state	SP	0.93	0.07
	AU	0.02	0.98
		Una.	Awa.
		Predicted label	

96%

AU–SP epochs correct

AP is absent from training and testing, so this validation cannot reveal sensitivity to paralysis.

The same classifiers fail on the awake-paralysed state

Use the same AU–SP classifier; add held-out AP epochs only at test time and label them “aware”.

LZ complexity

True state	SP	1.00	0.00
	AP	1.00	0.00
	AU	0.02	0.98
		Una.	Awa.
		Predicted label	

0%

AP epochs recognised as aware

Gamma power

True state	SP	0.96	0.04
	AP	0.99	0.01
	AU	0.00	1.00
		Una.	Awa.
		Predicted label	

1%

AP epochs recognised as aware

Alpha power

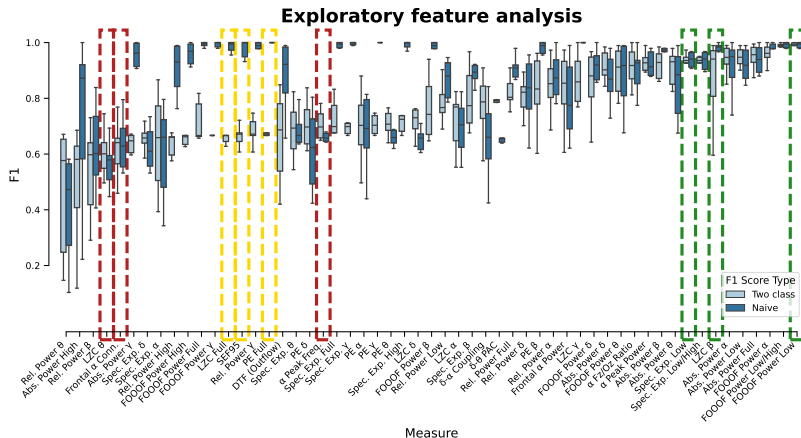
True state	SP	0.93	0.07
	AP	0.07	0.93
	AU	0.02	0.98
		Una.	Awa.
		Predicted label	

93%

AP epochs recognised as aware

95% CIs for AP accuracy: LZc [0.00–0.39], gamma [0.00–0.39], alpha [0.44–0.97].

Most candidate measures mistake paralysis for unawareness



Two bars per measure

dark = naïve (AU-SP only)
light = two-class (AP also tested as aware)

Boxed groups

red can't separate any state

LZc θ , frontal α conn., α peak freq.

yellow muscle-sensitive: big naïve $\rightarrow$ two-class drop

LZc full, SEF95, PE full

green best in both tests

FOOOF low, LZc β , spec. exp. low/high

Promising here does not yet mean clinically reliable

- ▶ The sample is small ($n = 6$) and reflects an exceptionally demanding experimental protocol.
- ▶ The sedated condition was comparable to deep-sedation, not surgical anaesthesia; unawareness could not be objectively verified (IFT removed).
- ▶ Alpha responses vary with anaesthetic agent, dose, age and clinical context.
- ▶ The findings require prospective replication in larger, clinically representative cohorts.

The awake-paralysed condition must become a validation standard

1. **Test motor state and conscious state independently.**
2. **Awake vs sedated accuracy may overestimate utility.**
3. **Treat high-frequency and complexity features cautiously.**

Take-home message

Near-perfect awake-sedated classification is not evidence of awareness detection if the same measure fails when an awake person is paralysed.

Acknowledgements

Co-authors

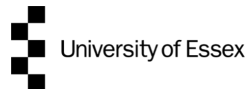
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Experimental team

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Reference

Halder, S., Juel, B. E., Pope, K. J., Hardy, A., Willoughby, J. O., and Storm, J. F. (2026).
An experimental study of the effect of neuromuscular blockade on EEG-based measures of awareness.
Scientific Reports. Nature Publishing Group.

- ▶ Looking for a PostDoc for start in March 2027
- ▶ Detection of pain from the EEG recorded during clinical procedures
- ▶ EEG analysis and machine learning (Deep Learning, Riemannian Classifiers) skills required
- ▶ Located in England
- ▶ Travel to collaborators in Norway and for conferences included!
- ▶ 18 Months duration
- ▶ Lots of interesting data!